11. $f(x) = \begin{cases} 1 + x & \text{if } x < -1 \\ x^2 & \text{if } -1 \leqslant x < 1 \\ 2 - x & \text{if } x \geqslant 1 \end{cases}$

15. $\lim\limits_{x \to 0^-} f(x) = -1, \quad \lim\limits_{x \to 0^+} f(x) = 2, \quad f(0) = 1$

16. $\lim\limits_{x \to 0} f(x) = 1, \quad \lim\limits_{x \to 3^-} f(x) = -2, \quad \lim\limits_{x \to 3^+} f(x) = 2,$

$f(0) = -1, \quad f(3) = 1$

REVIEW : TANGENTS, LIMITS & LIMIT LAWS

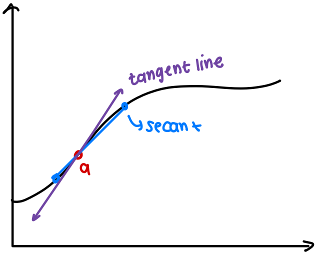
BIG PICTURE

We started w/ functions and the idea of average rate of change.

Then we looked closer & closer @ a point a

As x gets closer and closer to a the slope of the secant line approaches a single #.

That limiting slope is the **slope of the tangent line.**



Tangent line slope = $\lim_{x\to a} \frac{f(x)-f(a)}{x-a}$

↳ we call this instantaneous rate of change

AVERAGE VS INSTANTANEOUS RATE OF CHANGE

Average Rate of Change (over an interval)

$$\frac{f(b)-f(a)}{b-a}$$

- use 2 points a & b
- Tells us average slope between them

Instantaneous Rate of Change (at a single pt)

$$\lim_{x\to a} \frac{f(x)-f(a)}{x-a} = f'(a)$$

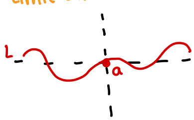
- looks @ values of x very close to a
- Gives slope of tangent line

LIMIT (INTUITIVE IDEA)

We say $\lim_{x\to a} f(x) = L$ if we can make f(x) as close as we want L by choosing x close enough to a (but not equal to a)

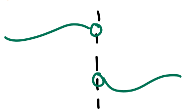
LIMITS FROM GRAPHS

Limit Exist & Equals L



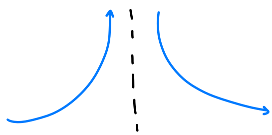
The left & Right sides both approach L

Limit Does NOT exist (JUMP)



Left & right sides approach different numbers.

LIMIT DOES NOT EXIST (INFINITE)



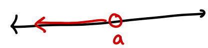
The function grows w/out bound near a.

ONE SIDED LIMITS

LEFT HAND LIMITS:

$\lim_{x\to a^-} f(x)$

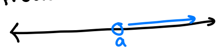
(approach a from the left)



RIGHT HAND LIMITS:

$\lim_{x\to a^+} f(x)$

(approach a from the right)



KEY RELATIONSHIP

- If $\lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = L$

  then $\lim_{x\to a} f(x) = L$

- If they are not equal the two-sided limit does not exist.

DAY 4 – LIMIT LAWS

LECTURE GOALS:

- Understood the laws of limits and why they work
- Use limit laws to evaluate limits algebraically
- Recognize important special limits
- Practice applying these ideas to many examples.

Why limit laws?

Limit laws let us break complicated limits into simpler parts that we already know how to find

They are the algebra rules that limits obey

LIMIT LAWS

Let $\lim_{x\to a} f(x) = L$ and $\lim_{x\to a} g(x) = M$ (L & M are real numbers)

1 CONSTANT LAW

$$\lim_{x\to a} c = c$$

2 Sum law

$$\lim_{x\to a} [f(x)-g(x)] = L+M$$

3 DIFFERENCE LAW

$$\lim_{x\to a} [f(x)-g(x)] = L-M$$

4 PRODUCT LAW

$$\lim_{x\to a} [f(x)g(x)] = LM$$

5 QUOTIENT LAW

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{L}{M} \quad \text{if } M \neq 0$$

6 POWER LAW

$$\lim_{x\to a} [f(x)]^n = L^n \quad (n \text{ an integer})$$

$$\lim_{x\to a} \sqrt[n]{f(x)} = \sqrt[n]{L}$$

$L \geq 0$ (not neg) when in ever

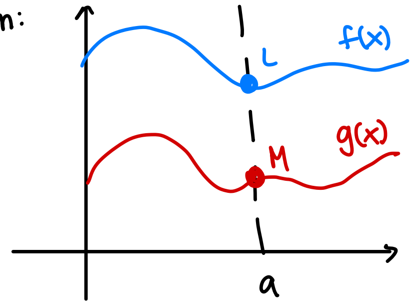
NOTE

- These laws only work when both individual limits exists & are finite
- For quotient law we must have $M \neq 0$
- For POWER LAW w/ even roots $L \geq 0$

WHY DO LIMITS LAWS WORK

Intuitively limits behave like numbers when we get close to a
If $f(x)$ gets close to L and $g(x)$ gets close to M then:

- their sum gets close to L+M
- their products get close to LM
- their ratio gets close to L/M



EXAMPLE: (SUM LAW)

If $\lim_{x\to a} f(x) = 3$ and $\lim_{x\to a} g(x) = -2$

what is $\lim_{x\to a} [f(x) + g(x)]$?

By the sum law:

$$\lim_{x\to a} [f(x) + g(x)] = 3 + (-2) = 1$$

EXAMPLE: QUOTIENT LAW

If $\lim_{x\to a} f(x) = 5$ & $\lim_{x\to a} g(x) = -3$

what $\lim_{x\to a} \frac{f(x)}{g(x)}$?

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{5}{-3} = -\frac{5}{3}$$

SPECIAL LIMITS

BASIC LIMITS

1) $\lim_{x\to a} c = c$
2) $\lim_{x\to a} x = a$
3) $\lim_{x\to a} kx = ka$
4) $\lim_{x\to a} x^n = a^n$ (n integer)
5) $\lim_{x\to a} \sqrt[n]{x} = \sqrt[n]{a}$ (if $a \geq 0$ n even)
6) $\lim_{x\to a} |x| = |a|$
7) $\lim_{x\to a} \sqrt{x} = \sqrt{a}$ (if $a \geq 0$)

TRIGONOMETRIC LAWS

1) $\lim_{x\to 0} \frac{\sin x}{x} = 1$ (x in radians)
2) $\lim_{x\to 0} \frac{\tan x}{x} = 1$
3) $\lim_{x\to 0} \frac{1-\cos x}{x} = 0$
4) $\lim_{x\to 0} \frac{\sin x}{1-\cos x} = 2$

Remember $\tan = \frac{\sin x}{\cos x}$

*each comes from trig identities

Derive:

$$\frac{\tan x}{x} = \frac{\sin x}{\cos x} \cdot \frac{1}{x}$$

$$\lim_{x\to 0} \frac{\sin x}{x} \cdot \frac{1}{\cos x} = 1 \cdot 1 = 1$$

HOW TO APPROACH A LIMIT PROBLEM

STEP 1: Try direct substitution first (plug in $x=a$)

STEP 2: IF substitution gives a real number you are usually done

STEP 3: If substitution gives $\frac{0}{0}$ do **NOT STOP** SIMPLIFY.

STEP 4: Look for algebra moves: factor, cancel common denominator, rationalize, identities.

STEP 5: If trig is involved, look for special limit patterns

STEP 6: Then apply limit laws again.

When direct substitution works

- If a function is a polynomial we can substitute the number directly
- If a function is a rational function $\frac{p(x)}{q(x)}$ & $q(a) \neq 0$ we can substitute a directly

EX POLYNOMIAL

Evaluate $\lim_{x \to 2} 3x^2 - 4x + 1$

Since this a polynomial substitute $x=2$

$$\lim_{x \to 2} 3x^2 - 4x + 1 = 3(2)^2 - 4(2) + 1$$
$$= 5$$

EX: ANOTHER POLYNOMIAL

Evaluate $\lim_{x \to -1} x^2 + 5x + 6$

Substitute $x = -1$

$$\lim_{x \to -1} x^2 + 5x + 6 = (-1)^2 + 5(-1) + 6$$
$$= 2$$

EX: Quotient

Evaluate $\lim_{x \to 3} \frac{x^2+1}{x-1}$

$f(x) = x^2 + 1 = 3^2 + 1 = 10$

$g(x) = x - 1 = 3 - 1 = 2$

$$\lim_{x \to 3} \frac{x^2+1}{x-1} = \frac{10}{2} = 5$$

What if substitution gives $\frac{0}{0}$?

If substituting gives $\frac{0}{0}$ this an indeterminate form. It means the limit might exist, but we can't tell yet

→ SIMPLIFY (factor, cancel, combine, etc).

then try again!

You try:

1) $\lim_{x \to 1} x^3 + 2$

2) $\lim_{x \to 0} 4x^2 - 7$

3) $\lim_{x \to 2} \frac{x^2 - 1}{x + 3}$

INDETERMINANT FORM

- When you substitute & get $\frac{0}{0}$ it does **NOT** mean the limit is 0 or undefined automatically
- It means the expression is indeterminate you must rewrite it first.
- Then use limit laws/special laws/ or direct sub.

EXAMPLE: FACTOR

Find $\lim_{x\to 3} \frac{x^2-9}{x-3}$ sub 3 = $\frac{3^2-9}{3-3} = \frac{0}{0}$

2) Factor $\frac{\cancel{(x-3)}(x+3)}{\cancel{x-3}} = x+3$

3) $\lim_{x\to 3} x+3 = 3+3 = 6$

EXAMPLE: RATIONALIZING

$\lim_{x\to 0} \frac{\sqrt{x+4}-2}{x}$

1) $x=0$ $\frac{\sqrt{0+4}-2}{0} = \frac{0}{0}$

2) multiply conjugate

$$= \frac{\sqrt{x+4}-2}{x} \cdot \frac{\sqrt{x+4}+2}{\sqrt{x+4}+2}$$

$$= \frac{(\sqrt{x+4})^2-4}{x(\sqrt{x+4}+2)} = \frac{x+\cancel{4}-\cancel{4}}{\cancel{x}(\sqrt{x+4}+2)}$$

$$= \frac{1}{\sqrt{x+4}+2}$$

3) limit

$$\lim_{x\to 0} \frac{1}{\sqrt{x+4}+2} = \frac{1}{\sqrt{4}+2} = \frac{1}{4}$$

COMMON ALGEBRA TOOLS

1) Factor & cancel common factors
2) Combine over a common denominator
3) Rationalize (multiply by conjugate)
4) Use trig identities

EXAMPLE: FACTOR

Find $\lim_{x\to 2} \frac{x^2-4}{x-2}$

1) Sub = $\frac{2^2-4}{2-2} = \frac{0}{0}$

2) Factor = $\frac{(x+2)\cancel{(x-2)}}{\cancel{(x-2)}} = x+2$

3) $\lim_{x\to 2} x+2 = 2+2 = 4$

EX: Algebra + special trig limit

$\lim_{x\to 0} \frac{1-\cos x}{x}$

1) sub $x=0$ $\frac{1-\cos 0}{0} = \frac{0}{0}$

2) $\lim_{x\to 0} \frac{1-\cos x}{x} = 0$ (special limit)

You Try:

1) $\lim_{x\to 4} \frac{x^2-16}{x-4}$

2) $\lim_{x\to 1} \frac{x^2-1}{x-1}$

3) $\lim_{x\to 0} \frac{\sqrt{x+1}-1}{x}$

SPECIAL LIMITS TRIG

1) MAIN SPECIAL LIMIT

1) $\lim_{x \to 0} \frac{\sin x}{x} = 1$

2) $\lim_{x \to 0} \frac{\tan x}{x} = 1$

3) $\lim_{x \to 0} \frac{1 - \cos x}{x} = 0$

How to recognize the pattern

- Look for $\frac{\sin(u)}{u}$ or expression that can be written to rematch them
- If the inside is not just x multiply & divide by the inside (or factor out constants) to create pattern Ex: $\frac{\sin 4x}{x} \Rightarrow \frac{\sin 4x}{4x}$ ✓ so mult $\frac{4}{4}$

Examples

1) $\lim_{x \to 0} \frac{5 \sin x}{x}$ plug in 0 $\frac{5\sin(0)}{0} = \frac{0}{0}$ ✗

$= \lim_{x \to 0} 5 \cdot \underbrace{\frac{\sin x}{x}}_{\text{SPECIAL LAW!}}$

$= \lim_{x \to 0} 5 \lim_{x \to 0} \frac{\sin x}{x}$

$= 5 \cdot 1$

$= 5$

2) $\lim_{x \to 0} \frac{\sin(4x)}{x}$ Plug 0 $\frac{\sin(0)}{0} = \frac{0}{0}$

$= \lim_{x \to 0} \frac{\sin 4x}{x} \cdot \frac{4}{4}$ Is this in $\frac{\sin x}{x}$ form

$= \lim_{x \to 0} \frac{\sin 4x}{4x} \cdot 4$ no! To make it $\frac{\sin 4x}{4x}$

$= 1 \cdot 4$

$= 4$

3) $\lim_{x \to 0} \frac{\tan(3x)}{5x}$ again we want → $\frac{\tan 3x}{3x}$

$\lim_{x \to 0} = \frac{\tan(3x)}{3x} \cdot \frac{3}{5}$

$= 1 \cdot \frac{3}{5} = \frac{3}{5}$

4) $\lim_{x \to 0} \frac{1 - \cos x}{x}$

$= 0$

You try: #1 $\lim_{x \to 0} \frac{7 \sin x}{x}$ #2 $\lim_{x \to 0} \frac{\tan x}{2x}$ #3 $\lim_{x \to 0} \frac{\sin(5x)}{x}$